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Pulley arrangement for a belt drive and agricultural machine with such

Abstract

A pulley arrangement for a belt drive including an input shaft having first end and second end. The first end of the input shaft is couplable to a gear box. Further, the pulley arrangement includes a pulley mounted coaxially on the input shaft having an inner pulley surface and an outer pulley surface. The pulley arrangement further includes at least one ring disc. The at least one ring disc is fastened coaxially to the inner pulley surface or the outer pulley surface. A plurality of air flow directing protrusions, grooves, or both protrusions and grooves extend from an inner diameter to an outer diameter of the ring disc, and a circumferential ring disc gap is provided between the at least one ring disc and the input shaft.

Inventors: Perez Ramirez; Javier J. (Pinto, ES), Vergara Martinez; Daniel (Torrejon de Ardoz, ES)

Applicant: DEERE & COMPANY (Moline, IL)

Family ID: 87280859

Assignee: DEERE & COMPANY (Moline, IL)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to European Patent Application No. 23185394.6, filed on Jul. 13, 2023, which is hereby incorporated by reference.

FIELD OF THE DISCLOSURE

(2) The disclosure relates to a pulley arrangement.

(3) The disclosure further relates to the pulley arrangement for a belt drive of an agricultural machine.

BACKGROUND

(4) Agricultural machines, such as combine harvesters and balers are well known and are used for

travel through fields for agricultural processes such as harvesting, collecting and bundling the crops etc.

SUMMARY

(5) During and/or after the agricultural processes, debris and dust may be produced. Debris such as chaff, stalks, cobs, etc. are for example the result of the chopping process within a harvesting machine. Other debris may be due to spreading of existing debris lying on the field during the travel of agricultural machines in the field. This debris and dust may accumulate on the components of the machine such as drives, gearboxes, pulleys etc. during its travel. The agricultural machines may be stopped due to the debris, resulting in undesirable downtime. The accumulation of the debris around a pulley arrangement for a belt drive and its components may lead to friction between pulley, its components and/or housing of the pulley. Further, it impacts the lubrication connections to the pulley arrangements.

(6) An arrangement to reduce or to avoid such debris accumulation on the pulley arrangement may be provided.

(7) The disclosure relates to a pulley arrangement including an input shaft, a pulley and at least one ring disc. The input shaft includes a first end and a second end. The first end of the input shaft is couplable to a gear box. The pulley, having an inner pulley surface and an outer pulley surface, is mounted coaxially on the input shaft.

(8) The object will be achieved by the teaching of one or more implementations disclosed herein.

(9) Accordingly, a pulley arrangement of above-mentioned type will be provided. The pulley arrangement further includes the at least one ring disc fastened coaxially to the inner pulley surface or the outer pulley surface. A plurality of air flow directing protrusions, grooves, or protrusions and grooves extend from an inner diameter to an outer diameter of the ring disc. A circumferential ring disc gap is provided between the at least one ring disc and the input shaft. Alternatively, the ring disc may fasten coaxially to the inner pulley surface or the outer pulley surface with the help of bolts, screws, rivets or any such similar fasteners or couplers. Alternatively, the ring disc and the pulley may be welded, casted or forged together.

(10) Further, the at least one ring disc may include one ring disc fastened coaxially to the inner pulley surface and a circumferential pulley gap is provided between the pulley and the input shaft. The mentioned pulley gap may provide an air flow channel from the outer pulley surface towards the inner pulley surface to guide air to the ring disc fastened on the inner pulley surface.

(11) The at least one ring disc may include a first ring disc fastened coaxially to the outer pulley surface and a second ring disc fastened coaxially to the inner pulley surface and a circumferential pulley gap is provided between the pulley and the input shaft. Due to this alternative, the pulley is provided with a ring disc fastened on each of its sides. Providing a ring disc on both sides of the pulley will provide air flow to an area surrounding both pulley surfaces. The first ring disc, second ring disc and the pulley may be welded, casted or forged together.

(12) The ring discs may be disposed such that the plurality of air flow directing means of the respective ring disc is facing the respective pulley surface. Alternatively, the plurality of air flow directing means of ring discs may be available on both faces of the ring discs.

(13) A plurality of fastening apertures may be located circumferentially between the inner and the outer diameter of the ring disc. The fastening apertures may be for example holes of any suitable shapes such as for the above-mentioned bolts, screws, rivets or any such similar fasteners or couplers.

(14) The plurality of air flow directing protrusions, grooves, or both protrusions and grooves may extend in a curved shape towards an opposite direction of the rotation of the pulley. Alternatively, or in addition, the air flow directing protrusions, grooves, or both protrusions and grooves may extend in radially linear shape towards the opposite direction of the rotation of the pulley. In addition, in order to create an air suction or air push in an axial direction of the pulley, the shape of the air flow directing protrusions, grooves, or both protrusions and grooves may comprise a twisted

shape such as propeller blades or ventilator blades or such similar profiles.

(15) The plurality of air flow directing protrusions, grooves, or both protrusions and grooves may extend in a curved shape towards a direction of the rotation of the pulley. Alternatively, or in addition, the air flow directing protrusions, grooves, or both protrusions and grooves may extend in radially linear shape towards the direction of the rotation of the pulley. In addition, in order to create an air suction or air push in an axial direction of the pulley, the shape of the air flow directing protrusions, grooves, or both protrusions and grooves may comprise a twisted shape such as propeller blades or ventilator blades or such similar profiles.

(16) The air flow directing protrusions, grooves, or both protrusions and grooves are formed on a surface between the inner and the outer diameter of the at least one ring disc. The protrusions may also include fins or wings or blades. The air flow directing protrusions, grooves, or both protrusions and grooves may be welded or printed on the ring disc or the ring disc may be casted or forged together with the air flow directing means.

(17) Further, in one implementation a belt drive might be provided including a pulley arrangement as proposed above.

(18) Furthermore, an agricultural machine may include a belt drive as mentioned above. For example, the agricultural machine may be a combine harvester, a forage harvester or a baler.

(19) The above and other features will become apparent from the following detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 shows a schematic view of an agricultural machine in a form of a combine harvester having a belt drive.

(3) FIG. 2 shows a schematic view of the belt drive according to FIG. 1.

(4) FIG. 3 shows a cross sectional view of one example for a pulley arrangement of the belt drive according to FIG. 2.

(5) FIG. 4 shows a schematic view of a protection device of the pulley arrangement according to FIG. 3.

(6) FIG. 5 shows a cross sectional view of another example for a pulley arrangement of the belt drive according to FIG. 2.

(7) FIG. 6 shows a schematic perspective view of the pulley arrangement according to FIG. 3.

(8) FIG. 7 shows a cross sectional view of yet another example for a pulley arrangement of the belt drive according to FIG. 2.

DETAILED DESCRIPTION

(9) The implementations disclosed in the above drawings and the following detailed description are not intended to be exhaustive or to limit the present disclosure to these implementations.

(10) Referring now to FIG. 1 and FIG. 2, an agricultural machine in form of a combine harvester **10** is shown as an example. The agricultural machine may be also a forage harvester or a baler. The combine harvester **10** includes a main frame **12** that provides a structure for mounting combine body **14** and supported above the ground on forward drive wheels **16** and rear wheels **18**.

Alternatively, the combine harvester **10** may be fully or partially supported on ground-engaging crawler tracks. Further, the combine harvester **10** includes an operator station **20** which houses a seat for an operator and several controls which may be manipulated by the operator in order to control operations of the combine harvester **10**. In operation, a forward mounted header **22** cut a crop material from a field and converges it to a feeder house **24** for transferring it to process further inside the combine body **14**.

(11) Further, the cut crop material is threshed, separated, cleaned and disposed by a threshing and separating unit **26**, a cleaning system **28** and residue disposal system **30** respectively. The threshing and separating unit **26**, the cleaning system **28** and the residue disposal system **30** are suitably housed within the combine body **14**. The mentioned various operations require power, which is provided by an engine **32** mounted on the main frame **12** and housed within the combine body **14**. Alternatively, an electric drive may act as a prime mover and also fulfil the power requirement for various operations. Additionally, the electric drive may be an auxiliary drive to assist the engine **32**.

(12) A power transfer means **34** transfers power from engine to various components of the combine harvester **10** to perform these operations. As shown in FIG. 1, the power transfer means **34** comprise a belt drive **36** which includes a pulley arrangement **38**. For example, in one such operation, power from engine **32** is provided to the residue disposal system **30** as shown in FIG. 1.

(13) Referring now to FIG. 3, there is shown the pulley arrangement **38** for the belt drive **36**. The pulley arrangement **38** includes an input shaft **40**. The input shaft **40** is having first end **42** and second end **44**. The first end **42** of the input shaft **40** is configured to be couplable to a gear box **46**. Further, the pulley arrangement **38** comprises a pulley **48**. The pulley **48** has an inner pulley surface **50** and an outer pulley surface **52**. The pulley **48** is mounted coaxially on the input shaft **40**. The pulley **48** may be manufactured with extensive variety of materials such as aluminum, brass, cast iron, plastic, stainless steel, steel, and zinc alloy etc. to ensure greater stability. Further, the pulley arrangement **38** includes a protection device **54** in form of a ring disc **56**. The ring disc **56** is fastened coaxially to the outer pulley surface **52**. The ring disc **56** is fastened coaxially to the outer pulley surface **52** with the help of screws **58**. Further, a circumferential ring disc gap **60** is provided between the at least one ring disc **56** and the input shaft **40**.

(14) As shown in FIG. 4, a plurality of fastening apertures **62** are located circumferentially between the inner diameter **64** and the outer diameter **66** of the ring disc **56**. The fastening apertures **62** may be holes for receiving the above-mentioned screws **58**, or any other suitable type of aperture for another suitable type of fastener. Furthermore, a plurality of air flow directing means **68** extends from the inner diameter **64** to the outer diameter **66** of the ring disc **56**. The plurality of air flow directing means **68** extends in a curved shape from the inner diameter **64** to the outer diameter **66** of the ring disc **56**. The direction of the curves of the flow directing means **68** can be both towards or against the direction of rotation of the pulley **48**. In both cases, circulation of air, respectively an air curtain is created when the ring disc **56** is rotating, as described in more detail below. With that an accumulation of debris or dust is restrained from plugging on the pulley surfaces **50**, **52**.

(15) Further, the air flow directing means **68** are protrusions formed on a surface between the inner diameter **64** and the outer diameter **66** of the ring disc **56**. Alternatively, the air flow directing means **68** may be grooves formed on the surface between the inner diameter **64** and the outer diameter **66** of the ring disc **56**. In some implementations, both protrusions and grooves may be employed. As one example, as illustrated in FIG. 4, the air flow directing means **68** may be protrusions **68** that define grooves **69** therebetween.

(16) In operation, the belt drive **36** drives the pulley **48** which in turn rotate the ring disc **56** fastened to the pulley **48**. This creates a pressure difference to draw air axially inwards through the circumferential ring disc gap **60** and pushes it radially outwards through the air flow directing means **68** to generate a curtain of air at the outer pulley surface **52**. The curtain of air restrains accumulation or entry of debris into the pulley **48** or into components of the agricultural machine **10**.

(17) Further, FIG. 5 shows an implementation having a pulley arrangement **70**. In the pulley arrangement **70**, the ring disc **56** is fastened coaxially to the inner pulley surface **50**. Further, one or more of a circumferential pulley gap **72** is provided between the pulley **48** and the input shaft **40**. For a proper pulley arrangement **70**, a shim **74** is fastened at the outer pulley surface **52**. The shim **74** is designed to provide one or more shim gaps **76** between the shim **74** and the input shaft **40**. The one or more shim gaps **76** can be for example a circumferential gap **76** between inner diameter

of the shim **74** and outer diameter of the input shaft **40**. Alternatively or additionally, the shim gaps **76** can be provided by recesses or cut outs **77** (as indicated in FIG. **5** and FIG. **6** as an additional example) along the thickness of an inner circumferential edge of the shim **74** in addition or instead of the circumferential gap **76**. As provided for the shim **74**, similar recesses or cut outs (not shown) along the thickness of an inner circumferential edge of the pulley **48** can be provided instead or in addition to the circumferential pulley gap **72**. The similar recesses or cut-outs are aligned to the recesses or cut outs **77** of the shim **74**. This shim gap **76** along with the circumferential pulley gap **72**, respectively the alternative recesses or cut outs on the shim **74** and the pulley **48**, create an axial channel for air to be drawn inwards (towards the inner pulley surface **50**) in case of suction and air is pushed radially outwards through the air flow directing means **68** to generate the curtain of air. This curtain of air is generated at outer pulley surface **52** as the radially pushed air flows from the inner pulley surface **50** to the outer pulley surface **52** through openings **78** provided between spokes **80** of the pulley **48** as shown in FIG. **6**. The curtain of air as described restrains accumulation or entry of debris from outer pulley surface **52** into the pulley **48** or into components of the agricultural machine **10**. Additionally, a portion of the air curtain is formed at the inner pulley surface **50** too which restrains accumulation or entry of debris into the pulley **48** or into components of the agricultural machine **10** at inner side of the pulley **48**.

(18) Further FIG. **7** shows an implementation having a pulley arrangement **82**. In the pulley arrangement **82**, a first ring disc **84** fastened coaxially to the outer pulley surface **52** and a second ring disc **86** fastened coaxially to the inner pulley surface **50**. In this arrangement **82**, the circumferential pulley gap **72** is provided between the pulley **48** and the input shaft **40**. As already described above, also in this implementation in addition to or instead of the pulley gap **72**, the recesses or cut outs can be provided. In this implementation as shown in FIG. **7**, the first ring disc **84** is designed in a way to provide the circumferential ring disc gap **60** between the first ring disc **84** and the input shaft **40**. As already described above for the shim **74**, also in this implementation in addition to or instead of the circumferential ring gap **60**, the recesses or cut outs can be provided. The circumferential ring disc gap **60** along with the circumferential pulley gap **72**, respectively the alternative recesses or cut outs on the first ring disc **84** and on the pulley **48**, create an axial channel for air to be drawn inwards (towards the inner pulley surface **50**) due to suction and air is pushed radially outwards through the air flow directing means **68** to generate the curtain of air. The ring discs **84**, **86** are disposed such that the plurality of air flow directing means **68** of the respective ring disc **84**, **86** is facing the respective pulley surface **50**, **52**. In this implementation, the effects of creating air flow, respectively providing a curtain of air, having the ring disc on the outer pulley surface and on the inner pulley surface are combined.

(19) The terminology used herein is for the purpose of describing example implementations and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that any use of the terms “has,” “includes,” “comprises,” or the like, in this specification, identifies the presence of stated features, integers, steps, operations, elements, and/or components, but does not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(20) Those having ordinary skill in the art will recognize that terms such as “above,” “below,” “upward,” “downward,” “top,” “bottom,” etc., are used descriptively for the drawings, and do not represent limitations on the scope of the present disclosure, as defined by the appended claims. Furthermore, the teachings may be described herein in terms of functional and/or logical block components or various processing steps, which may include any number of hardware, software, and/or firmware components configured to perform the specified functions.

(21) Terms of degree, such as “generally,” “substantially,” or “approximately” are understood by those having ordinary skill in the art to refer to reasonable ranges outside of a given value or orientation, for example, general tolerances or positional relationships associated with

manufacturing, assembly, and use of the described implementations.

(22) As used herein, “e.g.,” is utilized to non-exhaustively list examples and carries the same meaning as alternative illustrative phrases such as “including,” “including, but not limited to,” and “including without limitation.” Unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., “and”) and that are also preceded by the phrase “one or more of” or “at least one of” indicate configurations or arrangements that potentially include individual elements of the list, or any combination thereof. For example, “at least one of A, B, and C” or “one or more of A, B, and C” indicates the possibilities of only A, only B, only C, or any combination of two or more of A, B, and C (e.g., A and B; B and C; A and C; or A, B, and C).

(23) While the above describes example implementations of the present disclosure, these descriptions should not be viewed in a restrictive or limiting sense. Rather, there are several variations and modifications which may be made without departing from the scope of the appended claims.

Claims

1. A pulley arrangement for a belt drive comprising: an input shaft having a first end and a second end, the first end of the input shaft being configured to be couplable to a gear box; a pulley mounted coaxially on the input shaft, the pulley including an opening, the input shaft extending through the opening; an inner pulley surface; an outer pulley surface, the opening extending from the inner pulley surface to the outer pulley surface; and at least one ring disc fastened coaxially to the inner pulley surface or the outer pulley surface, the at least one ring disc defining a plurality of air flow directing protrusions, grooves, or both protrusions and grooves, each extending from an inner diameter towards an outer diameter of the ring disc, and wherein a circumferential ring disc gap is provided between the at least one ring disc and the input shaft, and wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves is disposed proximate to the opening.

2. The pulley arrangement for a belt drive of claim 1, wherein the at least one ring disc includes one ring disc fastened coaxially to the inner pulley surface and a circumferential pulley gap is provided between the pulley and the input shaft.

3. The pulley arrangement for a belt drive of claim 1, wherein the at least one ring disc includes a first ring disc fastened coaxially to the outer pulley surface and a second ring disc fastened coaxially to the inner pulley surface, and a circumferential pulley gap is provided between the pulley and the input shaft.

4. The pulley arrangement for a belt drive of claim 3, wherein the first and second ring discs are disposed such that the plurality of air flow directing protrusions, grooves, or both protrusions and grooves of the respective ring disc is facing the respective pulley surface.

5. The pulley arrangement for a belt drive of claim 1, further comprising a plurality of fastening apertures disposed circumferentially between the inner and the outer diameter of the ring disc.

6. The pulley arrangement for a belt drive of claim 1, wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves each extend in a curved shape towards an opposite direction of the rotation of the pulley.

7. The pulley arrangement for a belt drive of claim 1, wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves each extend in a curved shape towards a direction of rotation of the pulley.

8. The pulley arrangement for a belt drive of claim 1, wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves are formed on a surface between the inner and the outer diameter of the at least one ring disc.

9. A belt drive for an agricultural machine, the belt drive comprising: a pulley arrangement including an input shaft having a first end and a second end, the first end of the input shaft

configured to be couplable to a gear box, a pulley mounted coaxially on the input shaft, the pulley including an opening, the input shaft extending through the opening, an inner pulley surface, an outer pulley surface, the opening extending from the inner pulley surface to the outer pulley surface, and at least one ring disc fastened coaxially to the inner pulley surface or the outer pulley surface, the at least one ring disc defining a plurality of air flow directing protrusions, grooves, or both protrusions and grooves, each extending from an inner diameter towards an outer diameter of the ring disc, and wherein a circumferential ring disc gap is provided between the at least one ring disc and the input shaft, and wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves is disposed proximate to the opening.

10. The belt drive of claim 9, wherein the at least one ring disc includes one ring disc fastened coaxially to the inner pulley surface and a circumferential pulley gap is provided between the pulley and the input shaft.

11. The belt drive of claim 9, wherein the at least one ring disc includes a first ring disc fastened coaxially to the outer pulley surface and a second ring disc fastened coaxially to the inner pulley surface, and a circumferential pulley gap is provided between the pulley and the input shaft.

12. The belt drive of claim 11, wherein the first and second ring discs are disposed such that the plurality of air flow directing protrusions, grooves, or both protrusions and grooves of the respective ring disc is facing the respective pulley surface.

13. An agricultural machine, comprising: a belt drive including a pulley arrangement, the pulley arrangement including an input shaft having a first end and a second end, the first end of the input shaft configured to be couplable to a gear box, a pulley mounted coaxially on the input shaft, the pulley including an opening, the input shaft extending through the opening, an inner pulley surface, an outer pulley surface, the opening extending from the inner pulley surface to the outer pulley surface, and at least one ring disc fastened coaxially to the inner pulley surface or the outer pulley surface, the at least one ring disc defining a plurality of air flow directing protrusions, grooves, or both protrusions and grooves, each extending from an inner diameter towards an outer diameter of the ring disc, and wherein a circumferential ring disc gap is provided between the at least one ring disc and the input shaft, and wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves is disposed proximate to the opening.

14. The agricultural machine of claim 13, wherein the at least one ring disc includes one ring disc fastened coaxially to the inner pulley surface and a circumferential pulley gap is provided between the pulley and the input shaft.

15. The agricultural machine of claim 13, wherein the at least one ring disc includes a first ring disc fastened coaxially to the outer pulley surface and a second ring disc fastened coaxially to the inner pulley surface, and a circumferential pulley gap is provided between the pulley and the input shaft.

16. The agricultural machine of claim 15, wherein the first and second ring discs are disposed such that the plurality of air flow directing protrusions, grooves, or both protrusions and grooves of the respective ring disc is facing the respective pulley surface.

17. The agricultural machine of claim 13, further comprising a plurality of fastening apertures disposed circumferentially between the inner and the outer diameter of the ring disc.

18. The agricultural machine of claim 13, wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves each extend in a curved shape towards an opposite direction of the rotation of the pulley.

19. The agricultural machine of claim 13, wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves each extend in a curved shape towards a direction of rotation of the pulley.

20. The agricultural machine of claim 13, wherein the plurality of air flow directing protrusions, grooves, or both protrusions and grooves are formed on a surface between the inner and the outer diameter of the at least one ring disc.
